Which crime type is it? Inference and prediction for robbery among reported major property crimes in Toronto, 2014–2023

Kaiwen Yang

Abstract

Reported property-type crime in Toronto changed in composition after 2019: auto theft rose while robbery fell. We ask a narrow question: given that a major property-type crime was reported, which type is it? Using 159,056 Toronto Police incident records from 2014 to 2023, we contrast robbery with auto theft, break and enter, and theft over, and answer it twice. A binomial logistic regression with neighbourhood-cluster-robust intervals describes how the odds of robbery vary by hour, month, weekday, solar darkness and a post-2019 period: the post-2019 odds ratio is 0.55 [0.52, 0.58], and 28 of 41 time terms have intervals that exclude one. A five-rung prediction ladder, from time only to gradient-boosted trees, evaluated on held-out neighbourhoods and on a held-out later period, shows how little those time terms predict: under neighbourhood-grouped cross-validation time alone reaches a PR-AUC of 0.197 [0.179, 0.214] against a baseline of 0.161, whereas adding place (premises, location type and coordinates) lifts it to 0.531 [0.504, 0.555] and census features add 0.012 [0.003, 0.021]. Statistically clear time associations therefore coexist with weak prediction from time alone. Estimates describe the composition of reported incidents, not the risk of being robbed.

1 Introduction

Questions about robbery are often framed as when and where it happens. Police-reported data can answer a different and narrower question, and answering it carefully matters because two quite different goals share one dataset. *Inference* asks which features are associated with the outcome and by how much; *prediction* asks how well the outcome can be forecast for new cases (Shmueli, 2010; Breiman, 2001b). A feature can be strongly and reliably associated with an outcome and still help little in predicting it, and the reverse.

We state the task in words that the rest of the paper keeps: *given that a major property-type crime was reported, which type is it?* The outcome is whether a reported incident is a robbery rather than an auto theft, a break and enter or a theft over. This is $P(\text{robbery} \mid \text{reported incident})$. It is not the probability that a person is robbed at a place or time, which would need exposure data such as foot traffic or the number of people present, and it carries no causal reading. Reporting itself varies by crime type and neighbourhood (Skogan, 1975; Lum and Isaac, 2016).

The paper builds on two earlier course projects on Toronto Police open data, a robbery classifier and a logistic-regression analysis of robbery. We re-ran both, corrected their problems and put them on common ground. Our contributions are:

- a corrected binomial model for the odds of robbery, with hour, month and weekday as categorical terms, solar darkness, a post-2019 period term and neighbourhood-cluster-robust intervals;
- an ablation ladder of five models on identical rows, evaluated by neighbourhood-grouped cross-validation and by a temporal holdout, with bootstrap intervals for every score and every rung-to-rung difference; and
- a direct contrast between the two analyses on the same data: many time terms are statistically clear, yet time alone adds little to prediction.

2 Data and task definition

Sources. Incident records come from the Toronto Police Service open-data Major Crime Indicator (MCI) files for robbery, auto theft, break and enter, and theft over. Neighbourhood features come from a City of Toronto census neighbourhood profile joined by neighbourhood: population, household size, rental properties, median and average individual income, three ratios of workers to population or labour force, and the difference between median and average individual income. These features are fixed within a neighbourhood and over time.

Definition. The unit is one incident record. The positive class is robbery; the negative class is auto theft, break and enter or theft over. Assault is a Major Crime Indicator but is not a property crime and is not in the inputs; bicycle theft, homicide and shootings are excluded by rule. The window is occurrence years 2014 to 2023, since reporting in these files starts in 2014. Rows need a valid date, an hour of day, a premises type,

Table 1. Sample flow and class counts. Each removal is a row count; the steps add up to the unified table.

Step	Rows
Rows in the four Toronto Police files	174,993
Removed: occurrence outside 2014 to 2023, or invalid date	332
Removed: coordinates outside the Toronto box	2,409
Removed: exact duplicates within a category	12,472
Removed: events labelled robbery and another type	708
Rows in the unified table	159,072
Removed: no neighbourhood (NSA)	16
Rows used by the models	159,056
Model rows by class	
Robbery	25,657
Auto Theft	52,223
Break and Enter	69,108
Theft Over	12,068

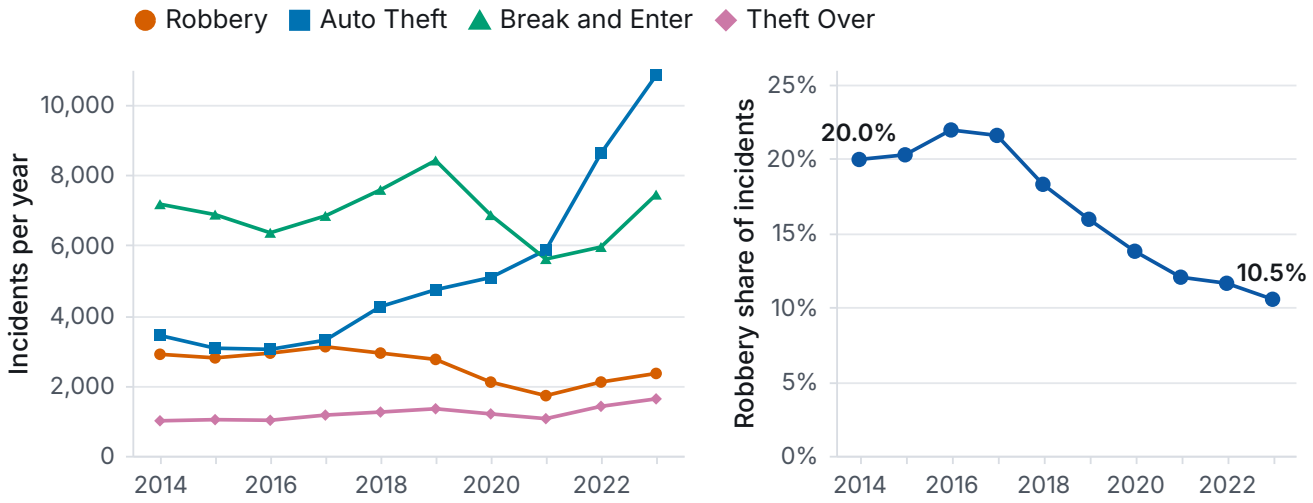


Figure 1. Incidents per year by crime type (left) and the share of incidents that are robberies (right), 2014–2023. Auto theft rises sharply after 2021, and robbery is lower in 2023 than in 2014, so the robbery share falls partly through the denominator.

coordinates inside the Toronto bounding box and a neighbourhood other than "not a neighbourhood"; exact duplicates within a category are removed. Events labelled both robbery and another type are removed (350 events), because their class is ambiguous. Table 1 gives the sample flow.

Composition shifts over time. The class mix is not stable (Figure 1). Robbery incidents fell from 2,893 in 2014 to 2,347 in 2023, while auto thefts rose from 5,082 in 2020 to 10,842 in 2023. Robbery's share of the four-crime total therefore fell from 20.0% to 10.5%, partly because the denominator grew. Any model of this outcome inherits that drift.

3 Methods

3.1 Inference: a binomial model of the odds of robbery

For incident i with outcome $y_i = 1$ if robbery, we fit a logistic regression with hour of day (reference: noon), month (reference: January), weekday (reference: Monday), an indicator that the sun was below civil twilight at the middle of the local hour, and an indicator for the post-2019 period (occurrence year 2020 or later). Darkness comes from NOAA solar equations for Toronto's coordinates, so unlike a fixed clock cut-off it follows the seasons; with hour and month as factors it is identified only by twilight hours that are dark on some dates and light on others. Hour and month are factors rather than linear numbers, and the factor coding fits better than a cyclic-harmonic alternative (AIC higher by 517 for the harmonic model). The model has 43 parameters and uses 159,056 rows. Incidents within a neighbourhood are not independent, so we report odds ratios with 95% intervals from cluster-robust standard errors (Cameron and Miller, 2015) over

158 neighbourhoods. As a discrimination check, a seeded stratified random five-fold cross-validated AUC is computed (it is not grouped by neighbourhood and includes the post-2019 term, so it is not comparable with the ladder's scores). The "post-2019" term absorbs every change after 2019 (reporting, policing, the mix of other crimes) and is not an effect of any single cause.

3.2 Prediction: an ablation ladder

Five models, called rungs, are fitted on identical rows, each adding one ingredient. **L1** uses time only: cyclic hour and month terms, weekday and darkness, in a regularised logistic regression. **L2** adds place: premises type, location type and a spline of the coordinates. **L3** adds the nine census features. **L4** replaces the logistic model with a random forest (Breiman, 2001a) and **L5** with gradient-boosted trees (XGBoost, Chen and Guestrin 2016). A constant-score baseline, L0, has ROC-AUC one half and a PR-AUC equal to the share of robberies. Because premises and location type nearly define some crime types (a break and enter happens in a building), every rung from L2 up is also fitted without them. Hyperparameters are chosen from small grids by PR-AUC in a three-fold neighbourhood-grouped cross-validation inside each training part.

3.3 Evaluation

We use two designs, each producing out-of-sample predictions for every held-out row. **Design A** is five-fold cross-validation grouped by neighbourhood, so each neighbourhood's census values and nearby coordinates never appear in both training and test parts (Roberts et al., 2017); it measures performance in new neighbourhoods. **Design B** trains on incidents through 2021 and tests on 2022 and 2023, the period when the robbery share was lower; it measures performance in later years. Models are scored on the pooled predictions by ROC-AUC, PR-AUC and F1, plus precision and recall, at the threshold that maximises F1 on inner out-of-fold predictions; the test rows never choose the threshold. PR-AUC depends on prevalence (Saito and Rehmsmeier, 2015), which differs between designs (16.1% and 11.0%), so values are compared only within a design. Intervals are percentile intervals from a cluster bootstrap over neighbourhoods (Efron and Tibshirani, 1993) with 1,000 replicates and a fixed seed; differences between rungs use the same resamples (paired). The importance of a feature block is the PR-AUC lost on the held-out rows when the block's columns are shuffled jointly, for the L5 model under Design A.

3.4 Corrections to the course versions

The two earlier analyses were re-run before being changed. Relative to them: the course regression's final summary table was accidentally a Gaussian linear model, so its coefficients were not odds ratios; hour, weekday and month entered as linear numbers, darkness was a fixed clock rule and standard errors ignored clustering; the reported AUC was apparent rather than cross-validated; the classifier used one random row split, so neighbourhoods appeared in both training and test parts, a fixed threshold of one half, and one year of data; and the written XGBoost tuning grid was never passed to the training call, so caret's default grid (108 combinations, shallow trees only) was searched instead of the intended 40. The versions here fix each of these, seed every random step and use one set of rows for every model.

4 Results

4.1 Inference: odds of robbery by time

Figure 2 shows the odds ratios. After 2019 the odds that a reported property-type incident is a robbery are about half what they were before: the post-2019 odds ratio is 0.55 (robust 95% interval 0.52 to 0.58). The composition check matters here. Fitting robbery against each other category separately gives 0.34 [0.31, 0.36] against auto theft, 0.81 [0.77, 0.86] against break and enter and 0.61 [0.57, 0.66] against theft over, so the pooled ratio mixes a fall in robbery with a large rise in auto theft, the comparison class. Time of day matters most. Against noon, the odds of robbery are highest at 15:00 (2.36, interval 2.12 to 2.63) and lowest at 08:00 (0.48, 0.40 to 0.57). After dark the odds are higher, with ratio 1.14 (1.06 to 1.22). Hour ratios are measured against noon, and the hours next to noon are themselves well above it (1.26 at 11:00 and 1.39 at 13:00), while the course-year counts spike at noon and midnight, which may reflect default or rounded times; we could not confirm a cause, so each ratio is a comparison with that one hour. Month and weekday terms are smaller: the largest month term is December against January (0.83, 0.78 to 0.90) and the largest weekday term is Saturday against Monday (1.13, 1.07 to 1.20). Of the 41 time terms, 28 have robust intervals that exclude one (we call these terms clear; no multiplicity adjustment is made). Even so, the model separates the classes only weakly: its cross-validated AUC is 0.631 (standard deviation 0.003 over folds).

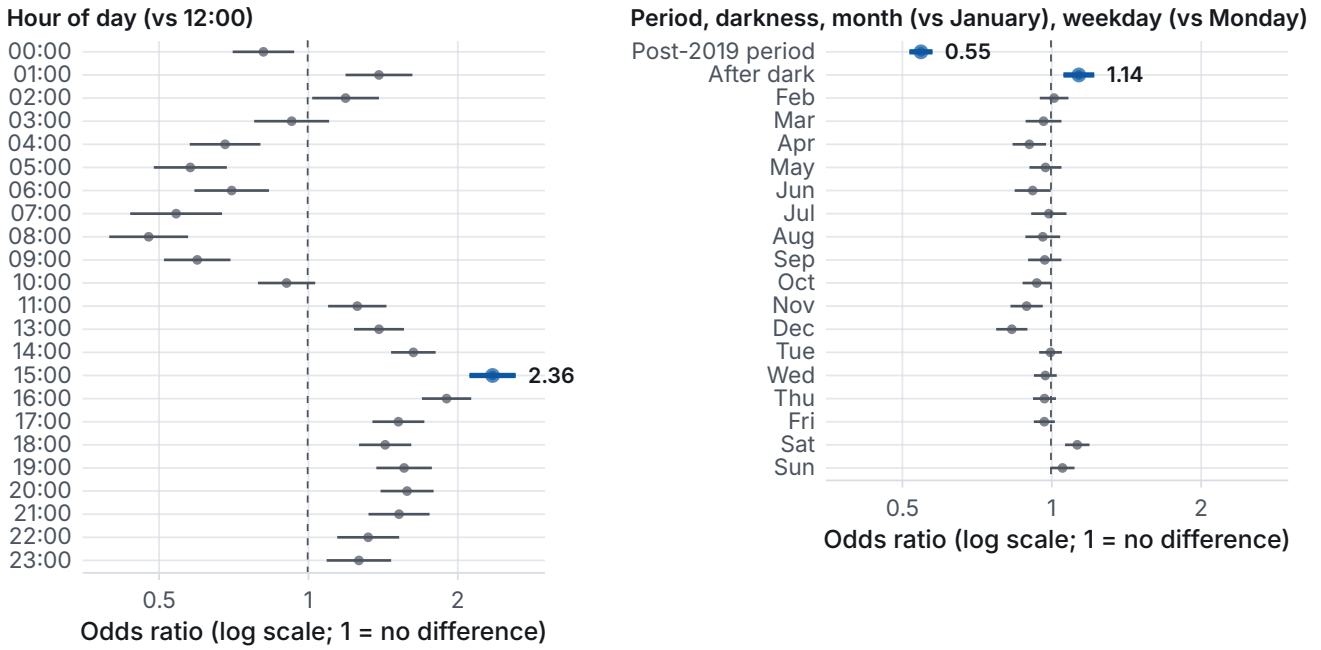


Figure 2. Odds ratios of robbery versus other major property crime, with cluster-robust 95% intervals on a log scale. Blue marks the post-2019 period, darkness and 15:00. Many hours and a few months have intervals that exclude one (the dashed line); most weekdays do not.

Table 2. Ladder metrics with 95% cluster-bootstrap intervals for every score. F1, precision and recall use a threshold tuned on inner out-of-fold predictions and are slightly optimistic. Prevalence differs between designs, so PR-AUC is compared only within a design.

Model	ROC-AUC	PR-AUC	F1	Precision	Recall
Neighbourhood-grouped 5-fold cross-validation					
L1 Time only	0.588 [0.579, 0.597]	0.197 [0.179, 0.214]	0.300 [0.278, 0.320]	0.183 [0.166, 0.198]	0.832 [0.823, 0.843]
L2 + place	0.836 [0.824, 0.847]	0.531 [0.504, 0.555]	0.546 [0.522, 0.565]	0.483 [0.448, 0.510]	0.628 [0.607, 0.647]
L3 + census	0.839 [0.830, 0.849]	0.543 [0.518, 0.566]	0.549 [0.530, 0.568]	0.490 [0.467, 0.512]	0.625 [0.604, 0.646]
L4 Random forest	0.858 [0.846, 0.869]	0.593 [0.561, 0.620]	0.562 [0.538, 0.583]	0.520 [0.486, 0.549]	0.611 [0.590, 0.631]
L5 XGBoost	0.860 [0.851, 0.869]	0.593 [0.564, 0.618]	0.564 [0.545, 0.582]	0.516 [0.493, 0.538]	0.622 [0.599, 0.645]
L2 + place, no premises	0.599 [0.576, 0.620]	0.202 [0.174, 0.232]	0.306 [0.282, 0.326]	0.194 [0.175, 0.210]	0.721 [0.681, 0.759]
L5 XGBoost, no premises	0.643 [0.626, 0.658]	0.242 [0.215, 0.273]	0.330 [0.303, 0.352]	0.220 [0.200, 0.237]	0.655 [0.603, 0.702]
Temporal holdout (train to 2021, test 2022 to 2023)					
L1 Time only	0.586 [0.574, 0.598]	0.141 [0.126, 0.157]	0.215 [0.195, 0.235]	0.125 [0.112, 0.138]	0.775 [0.760, 0.790]
L2 + place	0.808 [0.798, 0.819]	0.362 [0.336, 0.388]	0.385 [0.362, 0.407]	0.313 [0.290, 0.337]	0.499 [0.472, 0.525]
L3 + census	0.813 [0.802, 0.823]	0.377 [0.349, 0.404]	0.390 [0.366, 0.412]	0.318 [0.295, 0.342]	0.504 [0.478, 0.530]
L4 Random forest	0.840 [0.830, 0.849]	0.440 [0.411, 0.468]	0.438 [0.415, 0.459]	0.371 [0.349, 0.393]	0.533 [0.499, 0.563]
L5 XGBoost	0.841 [0.830, 0.849]	0.435 [0.406, 0.463]	0.437 [0.415, 0.458]	0.352 [0.331, 0.373]	0.575 [0.543, 0.606]
L2 + place, no premises	0.610 [0.593, 0.627]	0.158 [0.136, 0.184]	0.233 [0.211, 0.253]	0.141 [0.127, 0.155]	0.662 [0.612, 0.704]
L5 XGBoost, no premises	0.665 [0.642, 0.684]	0.202 [0.171, 0.234]	0.264 [0.246, 0.284]	0.168 [0.156, 0.181]	0.616 [0.552, 0.678]

4.2 Prediction: what each ingredient adds

Table 2 and Figure 3 give the ladder. Time alone is a weak predictor. Under Design A it reaches a PR-AUC of 0.197 [0.179, 0.214] against a baseline of 0.161, and a ROC-AUC of 0.588. Adding place (premises, location type and coordinates) is the big step: PR-AUC rises by 0.334 [0.318, 0.350] to 0.531. Nearly all of that step is premises and location type: removing them from L2 costs 0.328 of PR-AUC [0.308, 0.342]. The census block adds little once place is known: 0.012 of PR-AUC [0.003, 0.021] and 0.004 of ROC-AUC [−0.000, 0.008]. Non-linear models add more (random forest over L3: 0.050 of PR-AUC), but under Design A gradient boosting adds nothing measurable over the forest: the paired difference in PR-AUC has interval −0.005 to 0.005, which includes zero. Under Design B it is slightly worse (−0.010 to −0.001). Without premises and location type even the best model falls to a PR-AUC of 0.242.

The temporal holdout tells the same story at lower levels; ROC-AUC, which does not depend on the base rate, is lower too, and the robbery share has fallen to 11.0%. Time alone gives a PR-AUC of 0.141 [0.126, 0.157] against a baseline of 0.110, only modestly above it, and L5 reaches 0.435 (ROC-AUC 0.841 against 0.860 under Design A).

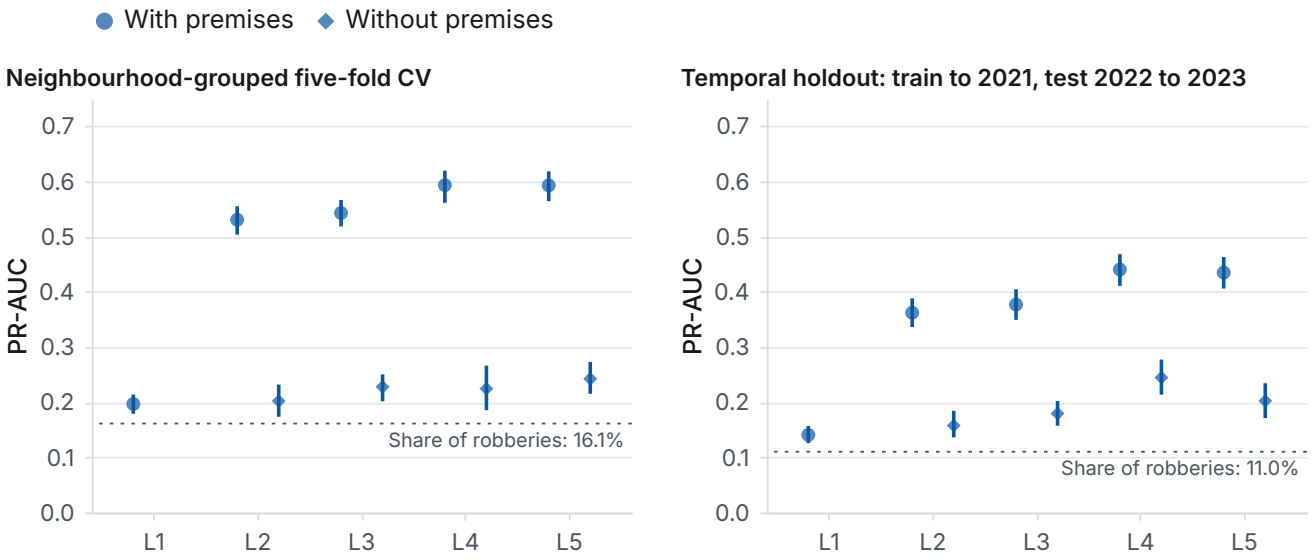


Figure 3. PR-AUC by rung (L1 time only; L2 adds place; L3 adds census; L4 random forest; L5 XGBoost) under neighbourhood-grouped cross-validation (left) and the temporal holdout (right), with 95% cluster-bootstrap intervals. Diamonds drop premises and location type. Dotted lines show the share of robberies, the PR-AUC of a constant score.

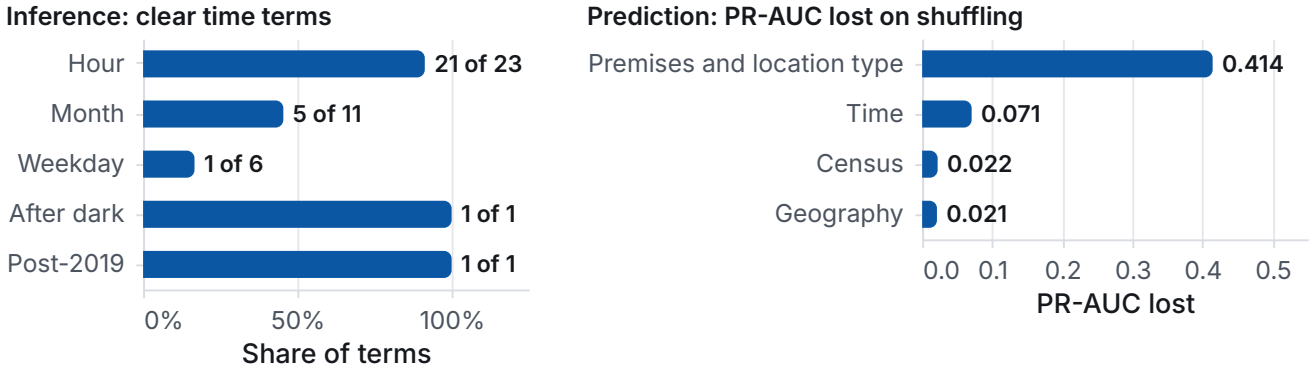


Figure 4. Left: for each time feature, the number of terms in the regression whose robust interval excludes one (clear terms) (labels give the count of clear terms out of all terms). Right: PR-AUC lost by the L5 model on held-out neighbourhoods when a block of features is shuffled; "Time" covers hour, month, weekday and darkness, and "Geography" is a smooth function of the coordinates.

4.3 Inference versus prediction

Figure 4 puts the two analyses side by side. In the regression, 21 of the 23 hour terms and every darkness and post-2019 term are statistically clear. In the prediction ladder, the time block matters much less than premises: shuffling it costs the L5 model 0.071 of PR-AUC under Design A, against 0.414 for premises and location type, and 0.022 for census and 0.021 for geography. Both findings are true at once. Some hour ratios are large, but they sit on a narrow range of predicted probabilities, so they sort few individual incidents; and with 159,056 rows even small associations get narrow intervals.

5 Discussion and limitations

Composition, not risk. The outcome is conditional on an incident having been reported as one of four crime types. A falling robbery odds ratio can come from fewer robberies, more of the other crimes or changes in reporting and classification; the data cannot separate these, and our composition check shows that the pooled post-2019 ratio depends heavily on the comparison class. None of the estimates is a probability of being robbed or an effect of time on crime.

Premises are nearly definitional. The largest predictive lift comes from premises and location type, which partly define the crime types (a break and enter occurs at a building). That lift is therefore not a discovery about robbery; it is why we report the no-premises ablation. Without those variables, even the best model has only modest predictive value.

Census features are coarse. The census block is a single profile that is constant within a neighbourhood and over the ten years, so it cannot track change within a neighbourhood and can proxy for neighbourhood identity. Grouped cross-validation prevents that from leaking across folds, and the block's small gain should be read with this in mind.

Data quality and shift. Police-recorded incidents reflect reporting and recording practices, which differ by crime type and neighbourhood (Skogan, 1975), and the published coordinates are offset by the publisher, so geography is approximate. The robbery share moved from 20.0% to 10.5%, a prior shift that lowers precision at a fixed threshold in the temporal holdout and means thresholds tuned in one period do not transfer unchanged. Because F1 thresholds are tuned on inner out-of-fold predictions from models that saw less data (and XGBoost was early-stopped on the scored fold), F1, precision and recall are slightly optimistic.

Significance at scale and scope. With 159,056 rows, intervals are narrow, and a statistically clear term need not be large or useful. We also limit the study to four crime types and do not model assault, bicycle theft, homicide or shootings. Predictive scores answer only which type a reported incident is, not where or when to expect incidents.

6 Conclusion

Answering the same narrow question by inference and by prediction gives complementary, not contradictory, results. The odds that a reported property-type incident is a robbery fell after 2019 and vary strongly by hour, yet the time features alone improve only modestly on the base rate; premises and location type, which partly define the crime types, supply most of the predictive lift, and census features add little. Reporting inference and prediction together, with intervals and out-of-sample designs, keeps each claim to what it supports.

Code and data availability

The code, the pinned checksums of the input files, all result files and a companion website with interactive versions of most of the figures (built with Vega-Lite, Satyanarayan et al. 2017) are in the project repository, `https://github.com/KaiwenYangUT/toronto-crime-analytics`. Every number in this paper is generated from the repository's result files. Incident data: Toronto Police Service Public Safety Data Portal, `https://data.tps.ca/`; contains information licensed under the Open Government Licence – Ontario. Neighbourhood features: City of Toronto Neighbourhood Profiles, `https://open.toronto.ca/dataset/neighbourhood-profiles/`; contains information licensed under the Open Government Licence – Toronto. This is an independent analysis and is not endorsed by the Toronto Police Service or the City of Toronto.

References

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001a.

Leo Breiman. Statistical modeling: The two cultures. *Statistical Science*, 16(3):199–231, 2001b.

A. Colin Cameron and Douglas L. Miller. A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372, 2015.

Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, 1993.

Kristian Lum and William Isaac. To predict and serve? *Significance*, 13(5):14–19, 2016.

David R. Roberts, Volker Bahn, Simone Ciuti, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017.

Takaya Saito and Marc Rehmsmeier. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3):e0118432, 2015.

Arvind Satyanarayan, Dominik Moritz, Kanit Wongsuphasawat, and Jeffrey Heer. Vega-lite: A grammar of interactive graphics. *IEEE Transactions on Visualization and Computer Graphics*, 23(1):341–350, 2017.

Galit Shmueli. To explain or to predict? *Statistical Science*, 25(3):289–310, 2010.

Wesley G. Skogan. Dimensions of the dark figure of unreported crime. National Criminal Justice Reference Service, NCJ 31968, 1975.